

 		VESSEL 22-V-1810 MECHANICAL SPECIFICATION				CLIENT IDENTIFICATION 22-3767-CD-SE-0V1810	
		Plant: 100kta Autoclave & 200kta Tubular LDPE/EVA	Project Name: MTO and Olefins downstream First-stage Project Jiangsu Sailboat Petrochemical Co		Location: Lianyungang – PR of China	TCM IDENTIFICATION CODE 3767-CD-SE-0V1810	Page 1 of 4

1	SERVICE	Expansion vessel				QUANTITY	1	
2	TYPE	VERTICAL	DIAMETER	mm	1800	PLANT UNIT	18	
3	HEAD TYPE	ELLIPTICAL (2:1)	HEIGHT or LEN.	mm	1700	INCLUDED IN		
4	INSULATION	Personal protection	CAPACITY	m³	5.5	SERVICE TYPE	CONTINUOUS	
5	INSTALLATION	Outdoor	FIRE PROOFING	NONE	RADIOGRAPHY	SPOT		
6	DESIGN CODE	ASME Section VIII Div.1 2010 Edition with 2011 Addenda + 3767-CZ-SG-000001						
7	OPERATING CONDITIONS							
8			VESSEL					
9	FLUID			Cooling medium				
10	FLUID HAZARDOUSNESS			000				
11	CONDITION 1	TEMPERATURE	°C	52				
12		PRESSURE	MPa[g]	0.60				
13		DENSITY	kg/m³	987.00				
14	CONDITION 2	TEMPERATURE	°C	155.0				
15		PRESSURE	MPa[g]	0.60				
16		DENSITY	kg/m³	911.00				
17	DESIGN CONDITIONS							
18			VESSEL					
19	MAXIMUM LIQUID DENSITY	kg/m³	1000.00					
20	CONDITION 1	TEMPERATURE	°C	170.0				
21		PRESSURE	MPa[g]	0.80				
22	CONDITION 2	TEMPERATURE	°C					
23		PRESSURE	MPa[g]					
24	FULL VACUUM	REQUIRED (yes/no)	Yes					
25		@ TEMPERATURE	°C	100				
26	METAL	MDMT	°C	-15.0				
27		@ PRESSURE	MPa[g]	0.85				
28	MATERIALS							
29			MATERIAL DESCRIPTION			CORROSION ALLOW.		
30	VESSEL	CS			mm	3		
31	INTERNALS				mm			
32	LINING				mm			
33	SUPPORTS	CS			mm	2 (For Skirt)		
34					mm			
35	GASKETS	VESSEL	According piping class					
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REVISED DATA	ROW NUMBER	DESCRIPTION			DRAWN UP	VERIFIED	APPROVED	DATE
	1	ISSUE FOR FINAL EPDD/BED			S. Uralath	N.Soni	C.Shah	06.11.2013
	0	ISSUE FOR EPDD / BED			W. Correia	N.Soni	C.Shah	23.09.2013

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							Page 2 of 4	ISSUE 1

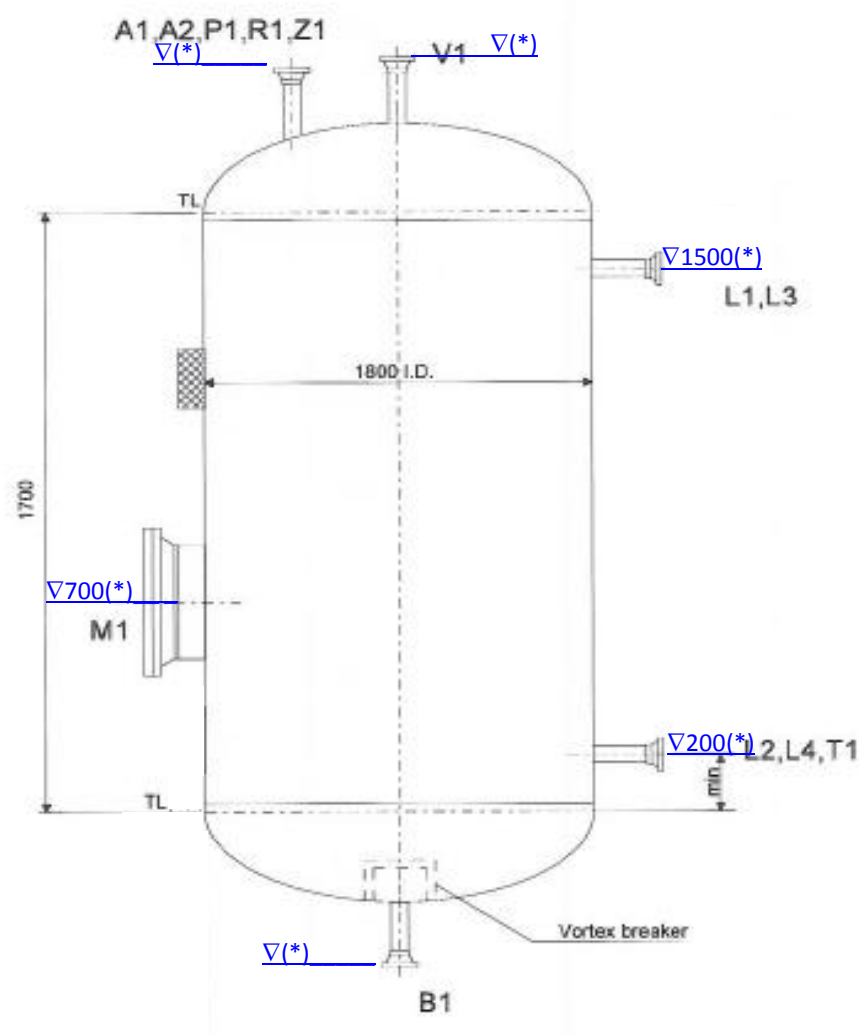
1	SERVICE Expansion vessel						QUANTITY	1
2							PLANT UNIT	18
3								
4	NOZZLES							
5								
6	POS.	N.	SERVICE	DN	RATING	FLANGE	FINISHING	NOTE
7	A1	1	INLET HW from header	2"	150 #	WN-RF		
8	A2	1	INLET Nitrogen from header	2"	150 #	WN-RF		
9	B1	1	OUTLET Water to 22-P-1810 A/B	10"	150 #	WN-RF		1*)
10	L1	1	Level instrument	2"	300 #	WN-RF		*3)
11	L2	1	Level instrument	2"	300 #	WN-RF		*3)
12	L3	1	Level instrument	2"	300 #	WN-RF		*2) *3)
13	L4	1	Level instrument	2"	300 #	WN-RF		*2) *3)
14	M1	1	MANHOLE	24"	150 #	WN-RF		w.bl.fl *4)
15	P1	1	Pressure instrument	2"	300 #	WN-RF		*3)
16	R1	1	RUPTURE DISK	(*)	300 #	WN-RF		
17	T1	1	Temperature instrument	1" 1/2	300 #	WN-RF		*3)
18	V1	1	VENT	2"	150 #	WN-RF		
19	Z1	1	SPARE	2"	150 #	WN-RF		w.bl.fl.
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REVISED DATA

ROW NUMBER

(*) = To Be Defined / Confirmed During Detailed Engineering

 		VESSEL 22-V-1810 MECHANICAL SPECIFICATION		CLIENT IDENTIFICATION 22-3767-CD-SE-0V1810	
				TCM IDENTIFICATION CODE 3767-CD-SE-0V1810	
		Plant: 100kta Autoclave & 200kta Tubular LDPE/EVA	Project Name: MTO and Olefins downstream First-stage Project Jiangsu Sailboat Petrochemical Co	Location: Lianyungang – PR of China	Page 3 of 4
					ISSUE 1
1	SERVICE	Expansion vessel			QUANTITY 1
2					PLANT UNIT 18
3					
4	NOTES				
5	GENERAL NOTES:				
6	Sheet 1, line 2: Diameter: I.D.				
7	Sheet 1, line 3: Height of cylindrical part.				
8	Sheet 1, line 4: Capacity to be confirmed by vendor.				
9	Sheet 1, line 11-16: CONDITION 1 refers to normal operating conditions.				
10	CONDITION 2 refers to refill operation with warm water.				
11	Sheet1, line 35: Gaskets for nozzle flanges shall be according to piping class.				
12					
13	NOZZLE NOTES:				
14	*1) With vortex breaker.				
15	*2) With reference vessel.				
16	*3) The size and rating of nozzle are to be specified by instrument department during detail engineering.				
17	*4) with davit				
18					
19	Mechanical Notes:				
20	1) PWHT - No				
21	2) All carbon steel material for pressure parts shall be killed and normalised.				
22	3) Manufacturer shall be responsible for calculation of all thicknesses for all conditions (pressure, wind and earthquake stability, etc) specified on this datasheet and attached specification.				
23	4) For Mechanical design, equipment shall be considered full of liquid.				
24	5) Painting upto finish coat is included in Manufacturer's scope. Only touch-up will be carried out in the field.				
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				Page 4 of 4	ISSUE 1
1	SERVICE Expansion vessel			QUANTITY	1
2				PLANT UNIT	18
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4	SKETCH (For Reference. Detailed Informaion to be provided during Detailed Engineering)				
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REVISED DATA	(*) = To Be Defined / Confirmed During Detailed Engineering NOTE: Type of support: Skirt, Skirt Length~ 2000mm & Support Base Elevation from ground: ~ 9050mm. To be Deifned/Confirmed during detail engineering.				
ROW NUMBER					